



Automated Waste Segregation & Gamified Recycling Optimization via YOLOv8

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INTRODUCTION & Data Engineering

- The Waste Crisis:** Traditional municipal waste management relies on manual sorting, which is highly inefficient, hazardous, and prone to cross-contamination. High-volume urban waste requires instant, automated sorting at the point of disposal.
- The ML Objective:** Deploy a high-throughput, edge-ready **YOLOv8** single-stage object detection framework that maps raw camera pixels directly to 2D bounding boxes and material classes in a single forward pass.
- Target Recyclables:** Systematically isolate municipal refuse into four clean, distinct material streams to foster a circular economy: **Glass, Metal, Paper, and Plastic**.
- The Gamification Layer:** To bridge deep learning with community action, the vision backend connects via a **FastAPI microservice** to **EcoTrack**—a smart mobile app built in **Flutter & Dart** that replaces tedious sorting with real-time point rewards, daily missions, and leaderboards.

METHODS AND MATERIALS

Dataset Architecture: 1,484 total images configured strictly in **YOLO Darknet format** paired with normalized spatial coordinate text files.

Preprocessing Infrastructure: Enforced local fast-cache staging to eliminate cloud I/O throttling. Integrated **441 negative background images** to minimize field false positives.

In-Flight Augmentations: Applied **Albumentations** (Blur, CLAHE), **Mosaic (factor 1.0)**, and **Mixup (factor 0.1)** to synthetically balance class distributions.

Core Engine: Deployed a 73-layer **YOLOv8 Nano** architecture (3,006,428 parameters) optimized via the **AdamW optimizer** over 20 epochs on a **T4 GPU**. Mosaic augmentations were deactivated for the final 10 epochs to refine bounding box regression accuracy.

RESULTS

- System Convergence:** Fine-tuned YOLOv8 Nano model optimized over 20 epochs on a T4 GPU. Bounding box and classification losses dropped smoothly to **0.4** with zero overfitting.
- Global Model Performance:** Overall Precision: **87.7%** | Recall: **73.4%** | mAP50: **81.4%** | mAP50-95: **71.1%**

Per-Class Performance Profiles

- Glass: 93.7% mAP50** balanced precision (83.1%) and recall (87.3%).
- Metal: 92.8% mAP50** excellent localization with an 85.6% mAP50-95 score.
- Plastic: 93.6% mAP50** robust feature mapping of crushed shapes yielding **90.5% recall**.
- Paper (Imbalance Bottleneck): 45.6% mAP50** high precision (92.4%), but limited recall (35.0%) due to severe training data scarcity (only 29 instances).



Figure 1: Raw Value Class Prediction Matrix.



Figure 2: Normalized Multi-Class Confusion Matrix..

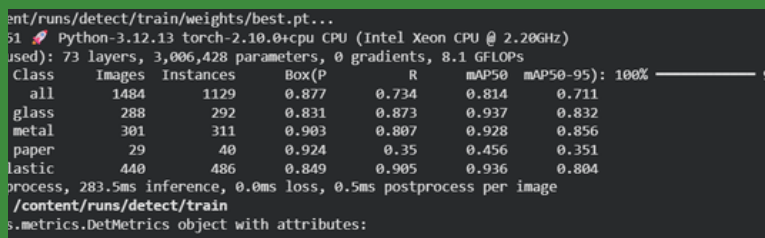


Figure 3: Training Loss Minimization and Accuracy Profiles.

EcoTrack Core System Integration

To bridge machine learning with real-world impact, the fine-tuned YOLOv8 model is deployed as a cloud microservice via **FastAPI** and linked directly to the **EcoTrack** mobile application ecosystem.

[User Camera Input] → [FastAPI Microservice Stream] → [YOLOv8 Inference] → [Global Leaderboard Sync] → [Point Allocation (+15)] → [JSON Class Payload]

- Frictionless Workflow:** The user scans an item through a native Flutter camera stream; the backend checks the image and returns a labeled bounding box with classification strings in **200–350 milliseconds**.
- Gamified Mechanics:** Validated items check off tasks in the **Daily Eco Missions Hub**, award points to the user's **Rewards Vault**, and instantly update rankings on the global **Eco Warriors Leaderboard**.
- Usability Fallback:** If confidence scores drop below 50%, a clean UI dropdown lets users manually select the material, ensuring absolute data integrity.

CONCLUSIONS

Outcome: Successfully deployed an edge-ready YOLOv8 sorting tool with a global **81.4% mAP50**.

Impact: Powering **EcoTrack** turns waste sorting into a rewarding daily habit.

Future: Integrate GANs for paper data scarcity and upgrade to Instance Segmentation for volume tracking.

REFERENCES

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ABSTRACT

Traditional municipal waste management relies heavily on manual segregation, which is labor-intensive, hazardous, and inefficient.

This project presents an automated, edge-deployable computer vision framework using **YOLOv8** to rapidly classify waste into four core recycling streams: glass, metal, paper, and plastic.

To encourage community participation, this system is integrated directly into **EcoTrack**, a gamified mobile platform built with **Flutter and Dart** that turns sustainable recycling into a daily habit through real-time point rewards and global leaderboards.

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